

UNIT-2: FUNDAMENTALS OF PATENT, TRANSFER AND INFRINGEMENT (9) Fundamentals of Patent: History of patents in India – Grant of patent – Inventions those are not patentable – Process and product patent – Specification and procedure of patent – e-filing – Temporal and spatial – Opposition to grant of patent – Rights and PCT of patents – Marketing rights – Milestones in Indian patent. Transfer and Infringement: Transfer and Infringement of patent rights – Surrender of patents – Challenges in patents

2.1 History of patents in India

Meaning of Patent : A patent is an exclusive right which is granted by the Government for an invention, for a limited time period.

This exclusive right of a patentee allows him to prevent a third party from using, selling or claiming the invention for which patent has been granted, in an unauthorised and illegal manner, while the term of the patent subsists. An invention which was granted patent can be used freely by the public after the term of the patent has expired or has ceased to have an effect.

Evolution of the patent system

The protection and enforcement of intellectual property rights should contribute to the promotion of technological innovation and to the transfer and dissemination of technology, to the mutual advantage of producers and users of technological knowledge and in a manner conducive to social and economic welfare, and to a balance of rights and obligations.

As a member-nation of the WTO, India was required to amend or enact laws to conform to the TRIPS Agreement. However, this was a challenge for India.

The journey of the Indian patent regime is reflected in three different periods: colonization, post-independence and globalization.⁷

Colonization. India inherited its patent regime from the British rule. When the British colonization of India ended, the Indian Patents and Designs Act, 1911, was in force and had created a system of patent administration in India under an administrative office – the Controller of Patents and Designs.

Post-independence. India enacted its first independent patent law in 1970. It focuses on the special socioeconomic conditions in India. Some of the significant changes introduced were with respect to food and drug patents, compulsory licensing, and connected working requirements. The law enacted in 1970 is credited with the growth of various industries, including the pharmaceutical industry, which, in two decades, gave India the distinction of being called “the pharmacy of the world” as Indian drug companies began exporting reasonably priced medicines to many countries.

Globalization. In 1991, India liberalized its economy and adhered to the General Agreement on Tariffs and Trade (GATT 1947), which was succeeded by the WTO, resulting in amendments being introduced in line with the TRIPS Agreement. These amendments saw India bring about fundamental changes permitting product patents in food, medicines and agrochemicals. The flexibilities in the TRIPS Agreement were used to maintain a balance: ensuring that the amendments would be gradually made systemic rather than forcing the closure of already-functioning industries. Statutory provisions relating to chemical and drug patents, patentability and other aspects of the amendments were tested repeatedly in the courts and were upheld as being within the Constitutional scheme while being fully compliant with the TRIPS Agreement. The judgment of the Supreme Court in *Novartis v. Union of India*⁸ recognized the need to curb the “evergreening” of patents while acknowledging the need to grant patent protection to incremental innovations. After *Novartis*, Indian courts have granted interim injunctions to protect patentees’ rights in pharmaceutical⁹ and agrochemical inventions.

The Indian Patents & Designs Act, 1911, was revised in 1950 after independence. The Indian Government created the **Justice N. Rajagopala Ayyangar Committee** in **1957** to look into the possibility of revising the famous Patent Law and make recommendations to the government. The patent legislation was established in 1970

2.2 Grant of patent

Grant of patents in India

Section 43 of the Patents Act, 1970 deals with the **Grant of patents**.

Conditions for Grant of Patents

The grant of a patent under the Patents Act, 1970 is not absolute. It is subject to certain conditions prescribed under **Section 47** of the Act. Therefore, the grant of a patent is subject to the following conditions:⁵

- the patent granted for any machine, apparatus, article or any article made by using a process, can be imported or made by the government or on its behalf solely for its own use;
- a patent granted for any process can be used by the Government or on its behalf for its own use;
- a patent granted for any machine, apparatus, article or any article made by using a process or where a patent is granted for any process, can be made or used by any person, for research or experiment purpose which also includes conveying instructions to students;
- where a patent is granted for any medicine or drug, the Government can import such drug or medicine solely for its own use or distribution in Government-supported dispensaries, hospitals, and other medical institutions and also those rendering public service.

Compliance of Conditions under the Act for Grant of Patent

A patent is granted promptly when:

- the patent application has not been declined by the Controller in the exercise of the powers vested in him; or
- the application is not in contravention of any provisions under the Act.

This includes within its scope:

- the objections raised by the examiner have been fulfilled and documents have been resubmitted that were returned with FER, within a period of 6 months or an extended period from the date of FER.
- where the FER refers to a prior art published before the date of filing of complete specification but after the date of priority, the applicant must prove that the priority date of the complete specification claimed is prior to the date on which the document was published. In the absence of such proof, the application can be refused.

Consequences of Grant of Patent

- every patent is assigned a serial number electronically, on being granted a patent. A “Certificate of Patent” is produced in a given format and an entry is made in the e-register. The date of recording the Patent in the Register of Patent and the date of grant of a patent by the Controller is the same.
- the complete specification of the invention which has been granted a patent is available on the official website.
- the public can inspect the patent application, specification, and other related documents by paying the requisite fee.
- the official journal of the Patent Office publishes the fact that the patent has been granted.
- the patentee must pay the requisite fee within 3 months on the grant of patent. This time period can be extended by 6 months.

- A post-grant opposition can be filed by any interested person within 12 months from the date of publication of the grant of patent.
- Every patentee must provide a statement with respect to the working of the invention to which a patent has been granted, commercially within a period of not less than 6 months, in a prescribed format.

2.3 Inventions those are not patentable

What Are Inventions?

As defined in **Section 2 (j)** the term “*invention* means a new product or process involving an inventive step and capable of application”. The invention should be of absolute novelty as neither it has been used nor published in any part of the world.

Section 3 And 4 Of The Indian Patent Act

Section 3 and Section 4 of the Patent Act is highly debatable and deals with the list of exclusions that are non-patentable that do not satisfy the above conditions. Following are not the “inventions” under the meaning of this act:

(a) Inventions that are frivolous

Inventions which are frivolous or contrary to well established natural laws.

Example– *Inventions that are against the natural laws that are any machine giving 100% efficiency, or any machine giving output without an input cannot be considered as obvious and cannot be patented.*

b) Inventions against public morality

Inventions in which the primary or intended use or commercial exploitation of which could be contrary to public order or morality (that is against the accepted norms of the society and is punishable as a crime) or which causes serious prejudice to human, animal or plant life or health or to the environment.

Example– *As in Biotechnology, termination of the germination of a seed by inserting a gene sequence that could lead to the disappearance of butterflies, any invention leading to theft or burglary, counterfeiting of currency notes, or bioterrorism.*

(c) Inventions that are a mere discovery of something that already exists in nature.

The mere discovery of a scientific principle or the formulation of an abstract theory or discovery of any living or non-living substances occurring in nature.

Explanation– *Mere discovery of something that is already existing freely in nature is a discovery and not an invention and hence cannot be patented unless it is used in the process of manufacturing an article or substance. For instance, the mere discovery of a micro-organism is not patentable.*

d) Mere adding of mixtures are non-patentable. A substance obtained by a mere admixing of two or more mixtures resulting only in the aggregation of the properties of the components thereof or a process for producing such substance is not considered the invention.

Explanation– *mere addition of mixtures is non-patentable unless this satisfies the requirement of synergistic effect i.e., interaction of two or more substances or agents to produce a combined effect greater than the separate effect.*

(e) Mere duplication of devices working in a known way is not an invention.

The mere aggregation or re-arrangement or duplication of known devices each functioning independently of one another in a known way.

Explanation– *mere improvement on something or combinations of different matters known before cannot be patentable unless this produces a new result or article.*

(f) Horticulture or agricultural method is non-patentable.

A method related to agriculture or horticulture.

Explanation– *a method of producing plants like cultivation of algae and mushrooms or improving the soil is not an invention and cannot be patentable.*

(g) Medicinal, curative, diagnostic methods for treating diseases in human and animals are non-patentable.

Any process for the medicinal, surgical, curative, diagnostic, or other treatment of human beings or any process for a similar treatment of animals to render them free of disease or to increase their economic value or that of their products.

Explanation– those medicinal methods administering medicines orally or injecting it, surgical methods like stitch free surgeries, curative methods as curing plaques etc does not fall under the ambit of the invention and are non- patentable.

Case law

In Mayo Collaborative Services V. Prometheus Laboratories, Inc²⁰.

In this case, the US Supreme Court said that “diagnostic and therapeutic methods (which includes the treatment or cure of diseases) is not patentable as it claims a law of nature”.

(h) Essential biological processes for the production of animals and plants is not an invention.

Plants and animals in whole or any part thereof other than micro-organisms but including seeds, varieties and species and essentially biological processes for production or propagation of plants and animals.

(i) Simple mathematical or business or computer programs are not an invention.

A mathematical or business method or a computer program per se or algorithms;

Explanation– *any mathematical calculation, any scientific truth or act of mental skills any activities related to business methods or algorithms (which are like the law of nature) cannot be patented.*

(j) Aesthetic creation is not an invention.

A literary, dramatic, musical or artistic work or any other aesthetic creation whatsoever including cinematographic works and television productions.

Explanation– *such activities like writings, painting, sculpting, choreographing, cinematographing all these which are related to creativity cannot be patented and fall under the gamut of Copyright Act, 1957.*

(k) Mental act, rule or method is not an invention.

A mere scheme or rule or method of performing mental act or method of playing a game.

Explanation- *playing a game such as chess, sudoku etc are not considered as inventions rather these are mere brain exercises and hence are not patented.*

(l) Traditional Knowledge is not an invention:

An invention which in effect, is traditional knowledge or which is an aggregation or duplication of known properties of the traditionally known component or components.

Explanation- *the traditional knowledge is know-how, skills, that is passed from generations to generations of a community and is already known cannot be patented for example the antiseptic properties of turmeric.*

(m) Atomic-Energy inventions are non -patentable.

Section 4 deals with inventions relating to atomic energy, that are also not patentable and that fall within sub-section (1) of section 20 of the Atomic Energy Act, 1962.

2.4 Process and product patent

Product patent vs process patent

The Indian Patents Act, 1970 provides for two types of patents, they are process and product patents. Let's discuss each of these types.

Product patent

As the name suggests, this type of patent protects the product. It offers the inventor higher protection for his invention by decreasing the level of competition of the same product. On the other hand, a process patent protects the manufacturing process of a product but not the product. The product patent maximises

the level of monopoly and minimizes the competition. So, we can conclude that a product patent has the following features that eventually benefits the inventor or the patent owner:

1. It provides a higher level of monopoly rights to the inventor of the patent owner.
2. Such a grant prevents others from manufacturing the same product using the same process or a different process.
3. Since the end product is given protection, the level of protection is considered higher in comparison with process patents.

Process patent

In this type, patent protection is granted only to a particular process used in manufacturing a product but not the end product. A process patent is often considered to provide limited protection. The reason is, it does not bar or prevent others from manufacturing or creating the same product by using a distinct process. Hence, it is possible that there are multiple process patents granted for a single product. This eventually reduces the monopoly that the inventor enjoys, thereby increasing the level of competition.

How are product and process patents different from each other

Sr.No	Basis	Product Patent	Process Patent
1	Definition	Patent protection is provided to the 'End Result' or 'the product'.	Process patent protection is provided to only the process, and not the resulting 'End Product'.
2	Competition	Once protection is granted, less competition	Competition shall remain
3	Monopoly	A higher level of monopoly is enjoyed by the inventor	Inventors do not enjoy a monopoly since other persons can still manufacture the same product using a different process.
4	Implementation	Product Patents were introduced as part of the Patents (Amendment) Act, 2005.	Process Patents have been recognised in India ever since the Indian Patent Act, 1970 was enforced.
5	Example	The altered DNA will be provided protection and not just the process involved.	The patent will be provided to the process involved in altering the DNA as recognised by the Indian Patents Act.

2.5 a Patent Specification

Patent Specification

A patent specification is a disclosure to the public at large regarding the invention as well as the scope of protection that would be granted to the invention. It provides an opportunity for the applicant to provide information regarding the invention in order to be entitled to claim protection. It is a crucial techno-legal document constituted by scientific and technical disclosures which designate the basis of the rights of a patent.

Section 7(4) of the Patents Act, 1970 (**the Act**) directs that every patent application shall be accompanied by a provisional or a complete specification. The Patent Rules, 2003 (**the Rules**) also particularly deal with the specification of an invention.

Structure of the specification

A patent specification normally has the following parts in the order given:

- A title to identify the invention.
- A statement as to the field to which the invention relates.
- An explanation of the background “state of the art” – what was already known prior to the invention.
- A statement of an objective technical problem which the invention seeks to solve, or an objective the invention is aimed at achieving.
- Statements of invention which correspond in scope and content to the claims, including any statements of technical advantages achieved by the invention.
- A list of any drawings and brief summary of what they show.
- A description of specific examples of the invention, often with reference to any drawings.
- The claims.
- An abstract.
- The drawings.
- A sequence listing (in cases involving nucleotide and amino acid sequences).

The title

This may be relatively uninformative if it is desired to maintain secrecy for as long as possible (the whole specification is usually published after 18 months, but the fact that an application has been filed, by whom and with what title, is usually available soon after filing)

The field of the invention

The start of the specification is a brief paragraph summarising the field in which the invention lies. This is usually framed as broadly as possible and helps the reader to appreciate the greatest extent of the invention as described and claimed in the rest of the specification.

The background art

This section mentions relevant “prior art” documents known to the applicant and summarises the state of the art in the field before the invention was made. The function of this section is to set the backdrop against which the invention is to be contrasted, thereby highlighting non-obvious technical contributions to the state of the art. Statements of objective technical problems arising in the art before the invention was made are of assistance.

Statement of the objective of the invention

Generally, this is simply a statement that the invention seeks to overcome or at least mitigate the problems identified with the prior art in the previous section. The stated objective should not be too ambitious.

Statements of invention

The objective should be achieved by the invention set out in this part. . These recite only the essential features of the invention. Then there are subsidiary statements of invention.

The list of drawings

This merely identifies the drawings used.

The claims

These set out the scope of the patent monopoly being sought by the applicant. Occasionally, with a first patent application, claims are not filed immediately, but they must be filed eventually.

The claims are a series of numbered paragraphs, each containing a single sentence. The first claim is independent but second and subsequent claims generally refer to and include the contents of the first (and perhaps subsequent claims) so that they are narrower in scope because they contain more limitations. There may be more than one independent claim.

Each independent claim serves two conflicting purposes. The first is that they must define something that is new and inventive compared with the prior art, otherwise no patent will be granted. They must have sufficient specific detail to distinguish from the prior art.

The claims must be clear and concise. The claims must also be supported technically by the description in the sense that the extent of the claimed invention must be reasonably credible from the specific description and the exemplification of the invention.

Abstract

If present, the purpose of the abstract is to assist in the official classification of the application on publication and to provide text by which third parties may locate the application when carrying out searches. The abstract may be brief and simply requires summarising in general terms what the specification is about.

The Drawings

These may be relatively crude so long as they are clear and show the required information. Known as “informal” drawings, they are used to keep initial filing costs under control. Replacement “formal” drawings, often prepared by a draughtsman, may be required in due course at additional cost in order to meet Patent Office requirements prior to publication of the application.

Please check that the drawings are correct technically and contain all of the necessary information. You should advise us of any corrections or additions required prior to filing.

Sequence listing

This is required in connection with biological inventions where nucleotide and/or amino acid sequences are present. It is presented as a text document with a particular format to enable patent offices to extract the information and facilitate searching of prior art documents during examination of this application, and of it, in due course, when it in turn constitutes prior art against subsequent applications.

2.5 b: Procedure of patent Filing and E- Filing

Eligibility for Patent Registration in India

Patent registration in India is subject to certain eligibility criteria that an invention must meet to qualify for protection.

1. **Novelty:** To be eligible for a patent, an invention must be novel, meaning it should not have been disclosed to the public before the date of filing the patent application. The invention should offer something new and inventive, and it should not be a part of the existing knowledge or prior art.
2. **Inventive Step:** An invention must possess an inventive step. The invention must involve a technical advancement or an inventive concept that is not readily deducible or obvious from existing knowledge or prior art.
3. **Industrial Applicability:** The invention must have industrial applicability, meaning it should be capable of being made or used in an industry. It should have practical utility and be applicable in any field of technology.
4. **Exclusions:** Not all inventions are eligible for patent registration. Certain subject matters are excluded from patent protection, including inventions that are contrary to public order, morality, or health. Discoveries, scientific theories, mathematical methods, aesthetic creations, and computer programs per se are also excluded from patentability.
5. **Ownership:** The person seeking patent registration must be the true and first inventor or the assignee of the inventor. In case of joint inventors, they should be listed as co-inventors in the patent application.
6. **Non-Disclosure to the Public:** It is important to note that the invention should not have been publicly disclosed before filing the patent application. Public disclosure may affect the novelty and non obviousness of the invention, potentially jeopardizing its eligibility for patent protection.

Stepwise Procedure for Patent Registration in India

Patent Registration process can be completed in multiple stages, beginning from Patent Search to the issuance of Patent Registration Certificate.

Step 1: Patent Search

Patent Search is the first step of Patent Filing procedure in India and is conducted worldwide to determine the novelty of an invention. Generally, it is considered safe to do patent searches before patent application filing. If an invention is found in prior arts or closely resembling prior arts, then the novelty of that invention can be challenged by the Indian Patent Office and the application can be objected to. Therefore, it is important to perform prior patent searches to assess the chances of your patent getting approved by the Patent Office.

Step 2: Drafting Patent Specification and Patent Application Filing

After conducting thorough searches worldwide, the specification of the invention is written in a technical language with or without the inventor's claims. Without claims is the provisional specification and with claims is the complete specification. The specification outlines the field of invention, detailed description of the invention with working examples, and the best method to use the invention so that when it comes to the knowledge of a person skilled in the art, it can be easily used. Legal protection to the patent is granted when the specification is drafted with the inventor's claim and is complete.

After drafting the Patent Specification, begin the procedure for Filing Patent Application in India. The drafted provisional or complete specification is filed in Form-2 while Patent Application form is filed in Form 1 as prescribed by the Indian Patent Act. If a provisional patent specification is filed, then within 12 months of its filing, a complete specification has to be filed with the inventor's claims. There are 6 different types of Patent Application forms which can be filed on the basis of their purpose. These are:

- **Ordinary patent application:** Used to file a patent application directly in India, seeking protection only within the Indian territory.

- **PCT National patent application:** Used to enter the national phase of PCT or Patent Cooperation Treaty in India, after which the applicant will be allowed to seek patent protection in multiple countries.
- **PCT International patent application:** Used to file a single patent application for patent registration in multiple countries which are members of the PCT, providing a centralized process for seeking patent protection internationally.
- **Convention patent application:** Used to claim priority based on an earlier application filed in a convention country, allowing applicants to secure the priority date for their invention.
- **Divisional patent application:** Used to divide an existing patent application into multiple separate applications, typically when the existing application contains multiple inventions.
- **Patent of addition application:** Used to file an application for an improvement or modification of an already existing invention for which a patent has already been granted.

Step 4: Patent Publication for Public Opposition

Once the patent application process is complete, then after the expiry of 18 months from the date of patent filing or date of priority whichever is earlier, the patent is published in an official journal and is open for public viewing and inspection. This provides an opportunity to the general public to raise an objection to the patent on valid grounds.

Step 5: Requesting Patent Examination

The patent application is examined only when a request for examination has been filed. The request for examination has to be filed within 48 months of the patent filing date or date of the priority. The patent examiner examines a patent application and issues an examination report. The examination report contains a series of objections raised by an examiner. The response to an examination report has to be filed within 12 months of the issuance of the examination report. If needed, the examiner can call the applicant or his agent for a show cause hearing to resolve the objections. That's why this phase is also called patent prosecution.

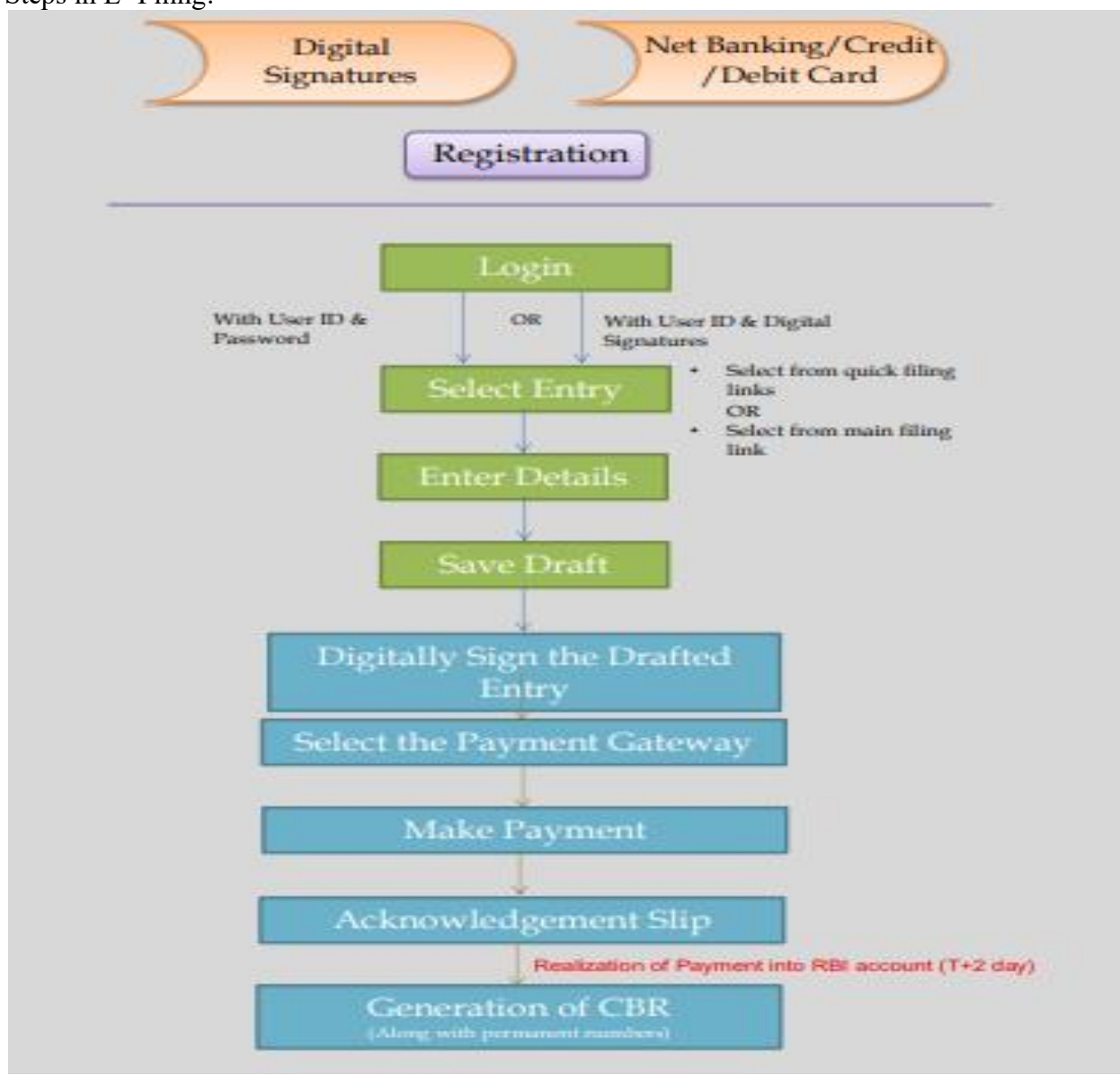
Step 6: Grant of a Patent

After all objections to the examination report have been replied to and the examiner is satisfied with the reply of the applicant, the application is put in order for grant of Patent Registration. This marks the end of the procedure for registration of Patent. On the other hand, if the examiner is not satisfied with the reply and arguments of the applicant, then he/she can reject the patent application. In this case, the applicant will again have to repeat the entire patent procedure in India again to get patent protection.

Cost & Validity of Patent Registration Applicant T	E-Filing Fee (INR)	Physical Filing Fee (INR)
Natural person(s) and/or Startup*	1,600 (within 30 pages, 10 claims)	1,750 (within 30 pages, 10 claims)
Small entity, alone or with natural person(s) and/or Startup	4,000 (within 30 pages, 10 claims)	4,400 (within 30 pages, 10 claims)
Others, alone or with natural person(s) and/or Startup* and/or small entity	8,000 (within 30 pages, 10 claims)	8,800 (within 30 pages, 10 claims)

Cost & Validity of Patent Registration Applicant T	E-Filing Fee (INR)	Physical Filing Fee (INR)
Additional Fees	160 per extra page 320 per extra claim	180 per extra page 350 per extra claim

Steps in E- Filing:



2.6 Temporal and spatial Aspects of Patent

The system of patent has temporal and spatial aspects. Temporal aspects relates to the time period and spatial aspects related to the geographical location.

FEATURES/DIFFERENCE BETWEEN TEMPORAL AND SPATIAL ASPECTS OF PATENT:

1, MEANING

TEMPORAL ASPECTS: It relates to the time period applicable for patents.

SPATIAL ASPECTS: It refers to the geographical location to which patent rights are exercised.

2. LIMITS

TEMPORAL ASPECTS: The patents are granted for a maximum period of 20 years.

SPATIAL ASPECTS: The patents are granted separately for each country. There is no patent called as world patent. The patent has to obtained I each country individually

3. ANALYSIS :

TEMPORAL ASPECTS: Analysis of patents over a time period which provides trends in patents during certain years.

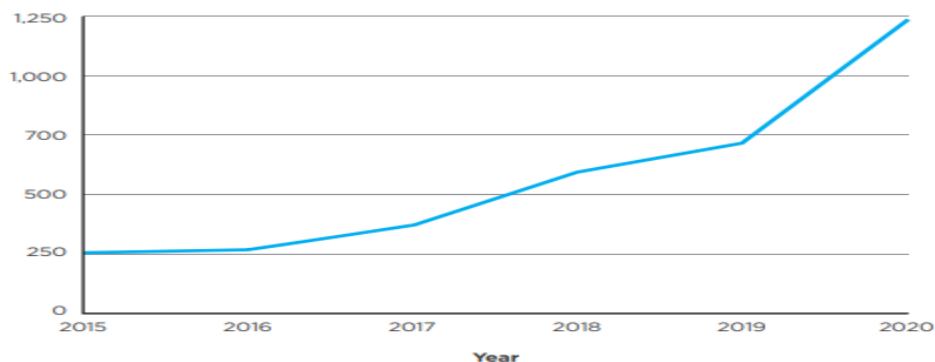
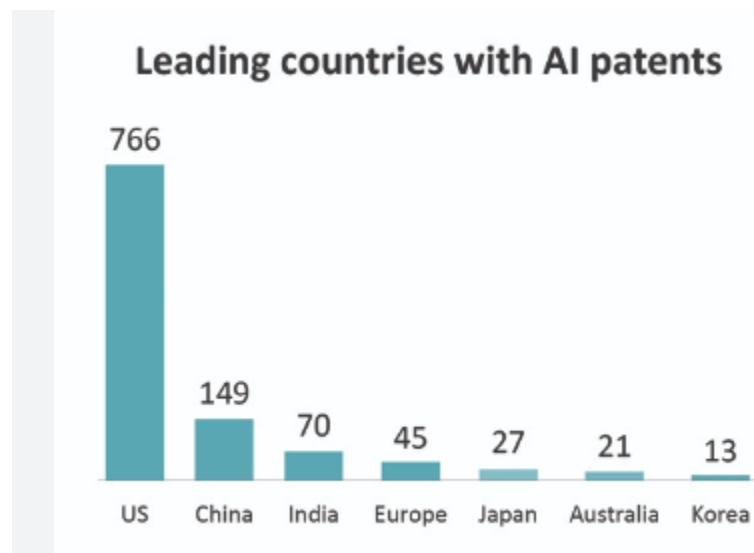


Fig. 5: AI-related publications on arXiv from India 2015- 2020

SPATIAL ASPECTS: The comparison of patents in different countries or region.



2.7 Opposition to grant of patent

Opposition In India: Complete Procedure

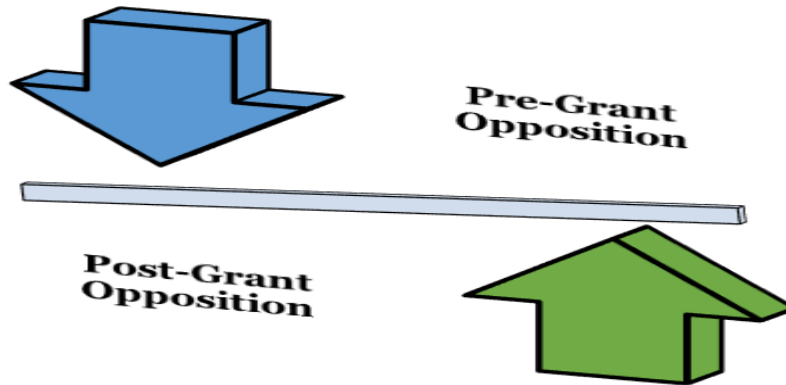
What Is Patent Opposition?

Patent Opposition proceedings provides opportunities for individuals or public at large or organizations to challenge the pending application of Patents or the granted Patents. The Patent Opposition, if appropriately used, can act as a vital tool in preventing the grant of frivolous Patents.

Even though the goal of preventing the grant of frivolous Patent can be attained by way of Revocation of Patent and other litigious options, but the Opposition of Patent serves as the most cost-effective route.

A Patent acquires a much higher Patent value if it passes the test of the Opposition of Patent. In India, the system of Patent is unique as it provides two kinds of process for the Opposition of Patent at two different phases of Patent prosecution.

The two types of Patent Opposition are as follows:



- Pre-Grant Opposition; and
- Post-Grant Opposition.

The grounds of Opposition of Patent were:

- Anticipation by publication which is prior;
- Prior Public knowledge and Prior public usage;
- Lack of inventive step and Obviousness;
- Non-patentability as per the Patent Act, 1970;
- **Inadequacy and clearness of description**

What Is Pre-Grant Patent Opposition?

Under Section 25(1) of the Patent Act, 1970, provides for the provision of filing a Pre-Grant Opposition of Patent against an application of Patent filed at IPO. As per Section 25(1), any person, any third party, or the Government can challenge the application filed for grant of Patent.

The opposing party should inform the Controller of Patents about the opposition, in writing, against the granting of the Patent. The Opposition of Patent should be made by the opponent party, after the publication of the application of Patent, but the Opposition should be filed before the grant of the Patent. Pre-grant Opposition of Patent serves as a defensive shield for confirming the validity of the applications of Patent before the granting of Patents to the applicants. The procedure of Pre-Grant Opposition of Patent acts as a security net to capture doubtful applications of Patent before a patent is granted to the applicants.

What Is The Procedure For Pre-Grant Patent Opposition?

The procedure for filing for Pre-Grant Opposition of Patent in India is as follows:

- The opponent has to file a representation for the Opposition in Form-7(A). The representation Form should be submitted along with the evidence and statement supporting the Opposition of

Patent. The representation can only be filed by the opponent, if a request for examination of an application of Patent is already filed with the IPO;

- The Controller of Patent after considering the representation decides that there is some merit in the representation filed by the opponent, he/she will give the notice of opposition of Patent to the applicant of the application of Patent;
- After the opposition notice is received by the applicant, he/she will file a reply to the representation of the opponent along with evidence supporting the application of Patent. The reply should be filed within 3 months from the date of receipt of the notice from Controller of Patent;
- The Controller, after considering the representation of opponent and reply of the applicant, along with the supporting evidences from both sides and after hearing both the parties, if a request for the same is made, can reject the Pre-Grant representation for Opposition of Patent and grant the Patent. On the other hand, the Controller, if is satisfied with the representation, can ask the applicant of Patent to amend the complete specifications of the Patent to his/her satisfaction before any further proceeding for the grant of Patent.

What Is Post-Grant Patent Opposition?

As per Section 25(2) of the Patents Act, 1970[1] any interested person can file for Post-Grant Opposition of Patent. The Opposition should be filed within 12 months from the date of publication of the grant of a Patent. The Opposition can be filed based on any of the grounds as prescribed in the Patent Act, 1970, by giving a notice of Opposition of Patent to the Controller of Patent.

The Post-Grant Opposition of Patent grounds is identical to those for the Pre-Grant Opposition of Patent. After notice is received, the Controller of the Patent should inform the Patentee about the Opposition of Patent and orders an Opposition Board to examine the Opposition made by the opponent and give its recommendations to the Controller.

What Is The Procedure For Post Grant Patent Opposition?

The procedure for Post-Grant Opposition of Patent is as follows:



Formation Of Opposition Board

The Controller of Patent will order the Opposition Board for the examination of the Opposition filed by the opponent. *The Opposition Board consists of a total of 3 members. Out of the 3 members of the Opposition Board, one member is nominated by the Controller of Patent to chair the Opposition Board.* Generally, the members of the Opposition Board are Patent examiners, except for that Patent examiner who has dealt with the same Patent application against which Opposition is filed.

Filing Of Documents

The opponent is required to file a copy of the written statement of Opposition and the supporting evidence of the Opposition.

In the written statement, the opponent must specify the following things:

- The nature of the interest of the opponent;
- The material facts on which the case is based;
- The relief which is sought; and
- any supporting evidences, if any.

Hearing

After the recommendations of the Opposition Board are received, the Controller of Patent should fix a date and time for hearing both the parties. The members of the Opposition Board are required to be present at the time of the hearing.

The Controller of Patent, after hearing both the parties, or without a hearing if neither of the party wishes to be heard, and considering the recommendations of Opposition Board, the Controller can give an order that the Patent be either maintained, amended or revoked.

What Are The Landmark Cases On Patent Opposition In India?

The landmarks cases on Patent opposition in India are as follows:

- ***Novartis AG vs. Natco Pharma Ltd***

Novartis AG, Switzerland filed an application for patent in India on 17th July, 1998, claiming Switzerland priority date of 18th July, 1997. Upon publication of the application, the grant of Patent was opposed by Natco Pharma Ltd., India company on 26th May, 2005.

2.8 a Rights of the Patentee

Patentee's Rights According to the Patent Act, 1970

The Patent Act of 1970 in India grants rights and obligations of patentees in India, allowing them to use their patented inventions or processes. The rights bestowed upon a patentee include the following:

1. Right to Exploit the Patent:

Under the rights and obligations of patentees in India, the first right ensues when the invention is a product. In this case, the patentee has the exclusive rights to use, make, import, or sell the invention within India. In the case of a process, the right to exploit refers to the exclusive authority to use or exercise the process within India.

2. Right to Grant licence:

The patentee holds the authority to grant licences, transfer rights, or establish agreements for consideration. For validity, any licence or assignment must be documented in writing and registered with the Controller of Patents. Unregistered assignments are not recognised as evidence of patent title.

3. Right to Surrender:

A patentee can choose to surrender their patent at any time by providing notice in the prescribed manner. This is usually done through publication in the Journal, allowing opposition if necessary. Surrender is exercised when the patentee anticipates non-performance of the patent in the future.

4. Right to Sue for Infringement:

Patentees have the right to initiate legal action against patent infringement in District Courts, which hold jurisdiction to adjudicate such cases.

5. Right to Use and Enjoy Patent:

The patentee exclusively enjoys the right to exercise, utilise, convey, offer, or practice the patented product or process within India.

Obligations of a Patentee Under the Patent Act, 1970

Upon being granted a patent, rights and obligations of patentees in India come into operation as per the Patent Act of 1970. These obligations ensure that the patent system operates effectively, encourages transparency, and maintains the balance between innovation and public interest. The patentee's obligations include the following aspects:

1. Duty to Disclose

Section 8 of the Patent Act, 1970, imposes the obligation of disclosure upon the patentee. This requires the patentee to furnish information regarding any corresponding or similar patent applications filed by them or on their behalf. Specifically, within 6 months from the filing date, the patentee is obligated to disclose these particulars. Additionally, ongoing updates about any subsequent applications must also be provided.

2. Duty to Request Examination (Section 11B):

Unlike some other intellectual property rights, the patent application process doesn't inherently include an automatic examination step. **Under Section 11B of the Patent Act, 1970**, the patentee has the responsibility to actively request the Controller of Patents to examine the application for patentability.

3. Duty to Respond to Objections:

After the examination request is submitted, the application undergoes scrutiny by a patent examiner. If objections arise during this evaluation, they are presented in the form of a First Examination Report. The patentee has the obligation to respond to these objections within a year from the issuance of the FER. Failure to comply could result in the application being deemed abandoned by the Controller of Patents.

4. Duty to Clear All Objections:

It's the patentee's responsibility not only to respond to raised objections but also to address and resolve them adequately. If necessary, the patentee may need to attend meetings with the Controller of Patents to discuss and resolve outstanding issues.

5. Duty to Pay Statutory Fees (Section 142):

The patentee is obligated to pay the stipulated statutory fees during the patent application process. Non-payment of these fees could lead to the non-processing or rejection of the application. **Section 142 of the Patent Act, 1970**, outlines the fee payment requirements and the potential consequences of non-compliance.

2.8 b PCT

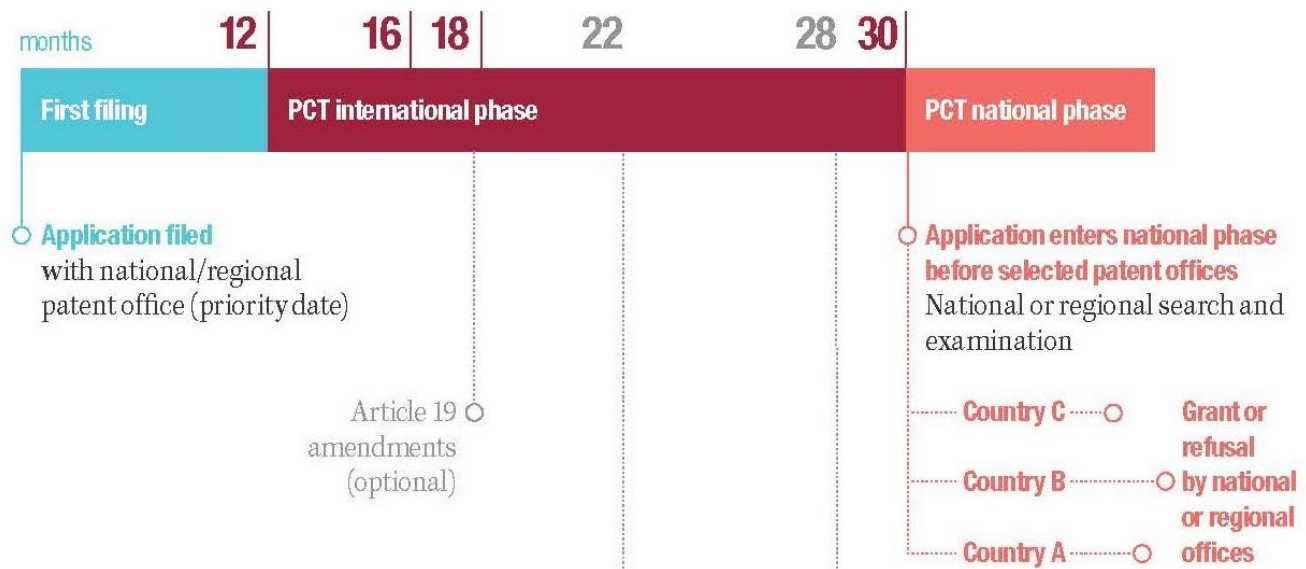
The Patent Cooperation Treaty (PCT) is an international IP agreement that provides patent protection in several countries through the filing of a single common application. The treaty, governed by the World Intellectual Property Organization (WIPO), has more than 150 nations as signatories, who are also known as PCT Contracting States.

Filing patent applications under PCT implies securing legal protection for an invention under patent law in more than 150 countries at once. Major global corporations, research institutes and universities seek patent protection via PCT. It safeguards applicants against accidental errors which are far more likely to occur if a separate application is filed in every country of interest.

The Patent Cooperation Treaty (PCT) assists applicants in seeking patent protection internationally for their inventions, helps patent offices with their patent granting decisions, and facilitates public access to a wealth of technical information relating to those inventions.

By filing one international patent application under the PCT, applicants can simultaneously seek protection for an invention in a large number of countries.

PCT ECHANISM/PROCESS:



PHASE 1 : NATIONAL PHASE/FIRST FILING:

First applicant need to file patent application in home country. Applicant need to file a national application along with international application in one language and need to pay single set of fee. PCT application is to be filed within **12 months** after filing a national application. Priority date is given by the national office.

PHASE 2 : INTERNATIONAL PHASE:

Application is transmitted to an International searching authority. An International Searching Authority search an application and check for a prior art and state of art. They made it clear whether or not document that have been presented is eligible for grant of patent. They establish it on the basis of their written opinion. Within **16 months** from the date of filing, ISR (International search report) will give search report with written opinion.

After submitting the necessary document user application is published, after the completion of **18 months**. WIPO publishes the PCT application. Content of application is disclosed to the general public.

PHASE 3 : NATIONAL PHASE:

After completion of all necessary requirements patent is granted by national office of different countries. National phase which follows the international phase in which rights are continued by filing necessary documents with the patent offices of separate countries of the PCT. Patent is granted to the applicant before the completion of **30 months**. After this applicant can seek protection.

2.9 Marketing Rights

Product Patent and Exclusive Marketing Rights-

The term Exclusive Marketing Rights (EMRs) means the right to sell or distribute the article or substance covered in a patent or patent application in the country. EMRs will be granted when there is no system of product patent in a country. It is only a temporary arrangement which will cease to have effect when product patent regime is introduced.

Exclusive Marketing Rights

EMR provision was introduced in India in the year 1999 in compliance with TRIPS as product patent for drug and medicine was not available in the Indian Act. As product patents can now be granted for Drugs, medicines, food, and chemical processes the EMR provision has become redundant and has been repealed.

Benefits of exclusive marketing rights

EMR provides an exclusive right to the person who files for a patent to distribute an output in their country for a period of five years. The applicant is also given the right to access to the agent, the sale or distribution of the article. While it is a provisional solution, it has its own benefits:

- It follows natural justice principle to the Owner and the public at the same time, as a useful and novel or new drug could be distributed while giving the due profits to the inventor.
- It is an effective process compared to the rigid terms of a patent that need to be satisfied.
- It is also in adherence to the norms of the TRIPs Agreement.

Article 70.9 of the agreement states that EMRs lay down the following conditions of obtaining an EMR for a particular product-

1. A patent application is pending in that member country where EMR is being sought.
2. A patent has been granted for that product in another member country.
3. Marketing approval has been obtained in such other member country.

In *Novartis AG v Adarsh Pharma and Anr. 2004*, the plaintiff was a Swiss pharmaceutical company that created a unique drug as a remedy for blood cancer. They had filed for patents in many countries where it was pending and they were also granted the patent for this drug in a few countries.

They contended that since India was a member of the WTO, it must adhere to its guidelines by providing them with EMR as an interim measure until their patent was granted.

It was held that the Plaintiff was eligible for EMR under Part IV-A of the Patent's Act and was thereby the court granted for the first time EMR in India.

2.10 : – Milestones in Indian patent.

Milestones in Patent:

- i. Amendments in Rules: The Patents Rules have been amended in 2016, 2017 and 2019 while the Trademarks Rules have been amended in 2017 to achieve the objective of removing procedural inconsistencies and unnecessary steps in processing of applications to speed up grant/registration and final disposal. By amending the Rules, the procedures are made more compact, time-bound, user-friendly and compatible for e-transactions. In Trademarks,

74 Existing Forms replaced by 8 consolidated Forms; One application form for all types of trademark applications provided; Process for determination of well-known mark laid out; Express provision for filing applications for sound marks provided; Procedures relating to registration as Registered User of trademarks simplified; and Expedited processing of an application right up to registration stage provided.

ii.. Use of IT and Technology:

- Paperless electronic processing implemented in Patent Office and Trade Marks Registry.
- E-filing of applications in Patents, Designs and Trademarks is available on 24x7 basis and has increased to more than 90% in Patents and Trademarks and 75% in Designs. 10% rebate on online filing in Patents and Trademarks to increase in online filing.
- Email as a mode of service has been implemented in all IPO transactions.
- Online delivery of Certificates of Grant/Registration of Patent, Trademark and Designs in digital format implemented.
- Registration of Patent Agents and issuance of certificate for registration of Patent Agents is made online.
- The office has started periodic publication of list of First Examination Reports (FER) issued by the Patent Office.

iii. Manpower Augmentation: In order to remove the backlog and enable speedy examination/disposal of IP applications, the manpower in IP Offices has been augmented substantially. Examiners of Patents & Designs have been recruited during 2016 and recruitment of 220 new examiners of patents in different fields was made during 2019 to have the sufficient strength.

iV India's and WIPO : The Government of India has accepted the proposal for India's accession to the WIPO Copyright Treaty and WIPO Performers and Phonograms Treaty, which extends the coverage of Copyright to the internet and digital environment, and also the proposal for accession of India which will help the Indian IP Office to harmonize the classification systems for examination of trademark and design applications, in line with the classification systems followed globally.

v. India and Japan: A pilot Patent Prosecution Highway (PPH) project with Japan has been signed and started in the month of December 2019

TRANSFER AND INFRINGEMENT:

2.11 Transfer

Transfer of Patent:

A patentee has the choice to prevent others from producing, using, practising, or selling the invention without his permission after receiving a patent.

A patentee can transfer the ownership of the patent to another to use the invention by way of:

- Assignment
- Licenses
- By activity of law

Assignment

Assignment is not defined in Indian Patents Act. The patentee transfers all or a part of his patent rights to the appointee who obtains the right to prevent others from creating, utilising, practising, or disseminating the invention. Section 50(3) of the Patents Act, 1970 states that in case a patent is co-owned by two or more individuals, any share of the patent cannot be assigned to anyone else without the consent of all co-owners.

Types of Assignment

There are three types of assignments.

- **Legal assignment**

An assignment (or consent to appoint) under a valid patent is one in which the trustee may identify himself as the patent owner. A deed must be used to allot a patent that was created through one. All patent rights are granted to a duly appointed person who is qualified to be its owner.

- **Equal or Equitable assignment**

Any form or agreement including a letter in which the proprietor agrees to share a certain share of the patent with another individual is referred to as the equitable assignment of the patent. In any case, an appointee in such a situation is not eligible to have his name listed as the patent owner in the register. However, they can notify the register of their interest in the patent.

- **Mortgages**

Mortgages are assignments in which the owner transfers all or a portion of their rights to the assignee in return for a fixed sum of money. The owner regains full ownership of the mortgaged property once the debt has been settled. The lender must have their name listed in the register as a mortgagee rather than registering themselves as the proprietor.

Licensing

Patent licensing is a process of granting permission to a third party to extract benefits by selling and using the licensed product. The patent owner gives license to a third party to use his patented invention based on the agreement and royalty. The license can be given for a period of time as per the mutual understanding between patent owner and licensee. During this time period, the licensee can use patented invention and can take financial benefits.

Types of License

The license may be exclusive or non exclusive. It is further classified into

- **Sub license**

Licensee has rights to issue Sub license to different organizations for making the product of patented invention. Patent owner give rights to licensee and the licensee has the right to issue the license further to a third party that can use patented invention. The financial benefits will depend on the contract between the primary licensee and third party.

- **Cross-license**

Cross-Licensing is the exchange of licenses between different organizations and creators. When invention requires the support of other products to make its place in the market, Cross-Licensing process is used.

- **Voluntary licensing**

Licensing on a voluntary basis is a gesture of goodwill towards the community. It also applies to patents for medicines. With voluntary licensing, the owner of a patent can grant the right to produce, import, or distribute a pharmaceutical product to other parties on an exclusive or non-exclusive basis. The licensee is allowed to sell and distribute the goods in a market, according the agreement. According to the terms and conditions outlined in the contract, the patent owner receives their royalty.

- **Compulsory licensing**

In Compulsory Licensing, the authorization is given to a third party to make, use or sell a patented invention without the consent of patent owner. According to the Sections 84 and 92 of Indian Patent Act 1970, if the specified conditions are satisfied, license can be granted to a third party without permission of patent owner. . Compulsory Licensing is usually reserved for pharmaceutical patents. Government allows someone to practice patented invention to make, use or sell patented invention without taking permission of patent owner for the public benefits.

- **Carrot licensing**

This strategy is appropriate when the potential licensee is not using the patented innovation and is not obligated to obtain a licence. In this situation, the patent holder must persuade the party to use his

product and explain how licencing it can be advantageous for them. In a marketing strategy known as carrot licencing, the owner of the patent tries to demonstrate to the licensee what may be accomplished by purchasing a licence for it.

- **Stick licensing**

Stick Licensing is another approach of licensing which is totally contrast of the carrot licensing. In Stick Licensing approach, prospective licensee is already using the patented technology and thus infringing the patent. The patent owner can file a suit against the infringer or settle with the infringer agreeing to license his patent.

- **Difference between Assignment and Licenses**

Assignment	License
Assignment refers to the transfer of ownership and of the patent to the assignee.	A license grants you the permission to use any patented invention, which would otherwise be referred to as infringement.
Assignments must be executed in writing.	Licenses can be granted without any written documentation.
Assignees don't need to pay any royalties to the original owner.	Licensees may be expected to pay certain royalties in exchange of using the inventor's product.
As you become a proprietor of the patent, it may be a bit expensive to get an assignment.	Licenses are much cheaper than assignments as you will also be paying some royalty in exchange of using the patent.

Transmission of Patent by Operation of law

When a patentee passes away, the patent's premium passes to his legal representative. If an organisation dissolves, goes out of business, or is liquidated, the patent is transferred by legal means.

2.12 Infringement of patent rights –

Patent Infringement:

Introduction

A patent is an invention-based legal document that defines and provides the bearer with exclusive rights to exclude others from producing, selling, or distributing such an invention. The violation of these exclusive rights of the patent holder is known as patent infringement. The patent is granted by the government for a limited period of time. This is to say that if the rights granted to the patentee are exercised by someone else who is not authorised by the patentee, then it would be considered a violation of patent rights, and the person is made liable for the same.

The Patents Act, 1970 does not provide specifications as to what would be considered an infringement of a patented product. But it lays down two kinds of activities (read with Section 48) which, when committed without the consent of the patent holder, would constitute infringement:

1. Making, using, offering for sale, selling, importing the patented product,

2. Using the patented process, or using, offering for sale, selling, or importing the product directly contained by that process.

Legislations governing patent infringement

The major enactment governing patent infringements in India is the Patents Act, 1970 which was implemented in 1972. It made pharmaceutical product innovations, as well as those for food and agrochemicals, un-patentable in India. It allowed for the copying and marketing of innovations in India that were patented elsewhere in the world. This Act also imposed restrictions on the import of finished formulas and introduced strict price control regulations. However, this act proved detrimental to foreign investments in the country since it did not benefit the big foreign multinational corporations and was not in consonance with the global patent system.

In 1992, India became a member of the WTO, and therefore, it became important to amend the existing law in order to meet the requirements of the TRIPS agreement. In order to comply with the TRIPS agreement, the introduction of Exclusive Marketing Rights (EMR) and the mailbox system was important. Under the EMR, exclusive rights would be provided to a foreign company to market a pharmaceutical or agricultural product in the Indian market for a specified period of 5 years.

Types of patent infringement

There are different kinds of patent infringement that are possible. They are listed and discussed as follows:

- Direct Infringement
- Indirect Infringement
- Wilful Infringement

Direct infringement

This is the most common type of patent infringement. As the name suggests, when a patented product or method (or substantially similar, i.e. equivalent to them) is used, marketed, sold, offered for sale, or imported without permission of the patentee during the term of such a patent, it constitutes direct patent infringement. It is considered to be of two types, viz- literal and non-literal patent infringement. They are discussed as follows:

Indirect or induced infringement

Indirect infringement is a type of patent infringement where the patentee's rights are involuntarily or unwillingly infringed by an infringer. It may so happen that some amount of deceit may be involved too. These days, products are becoming increasingly complex, and one end consumer product consists of multiple patented components. The manufacturer of such products cannot always have all the patent rights to the components of such products. Usually, the manufacturer takes licences for such required components to make and sell his products. But still, at times, a substantially similar product (mostly by accidents) is created because the indirect infringer had supplied some necessary components to make the substantially similar product, and the manufacturer ended up making the same without the full set of necessary permissions (licences) for each of the patented components. An illustration will make this clearer.

Illustration: Suppose X has a patent on a particular type of sound system. Y makes a substantially similar sound system (that infringes X's patent) with the help of Z's supply of a particularly essential component. Here, Y has committed direct infringement, and Z has committed indirect infringement of X's patent rights.

Willful infringement

As the name suggests, willful infringement is the type of patent infringement in which the infringer intentionally or willfully disregards and violates the patent rights of the patentee. In other words, if the infringer had knowledge of the patent and still violated the same, then such an infringement shall be a willful infringement.

So, what becomes important to establish here is that the infringer had knowledge of the patent, and the patentee has the burden to prove this to establish willful infringement. Usually, the patentee tries to discharge such onus by establishing that the infringer was duly served notice but continued infringement nonetheless. In such a case, the infringer has the defence to show that he had taken a legal opinion on the same and continued infringement because he believed in a *bona fide* manner that such a patent was either invalid or his actions did not constitute infringement.

If a patentee can successfully show willful infringement, then the infringer may have to face substantial pecuniary penalties that usually cover the legal fees of the patentee and even three to four times the actual damages faced by the plaintiff.

A significant case law in this regard is *Power Lift, Inc. v. Lang Tools, Inc. (1985)*, where the infringers (Lang Tools) were held to have infringed Power Lift's patent willfully.

Remedy for Patent infringement :

The patentee can file the case in the place of his residence or the place where he carries out his business, or where the cause of action arises. Section 48 of the Indian Patents Act contains the rights of the patentees. It lists the following activities as the infringement of the patentee's rights:

1. Using,
2. Making,
3. Importing,
4. Offering for sale patented products,
5. Selling the patented products or products directly obtained from the patented process.

The remedies available to the patentee are:

Temporary injunction

A temporary injunction is invoked by the court at the initial stages of the suit filed by the plaintiff. This is passed in order to prevent the defendant from getting further gains by using other patented products. In order to invoke a temporary injunction, it is important for the patentee to prove that the patent is valid and has been infringed by the defendant. Also, the subsequent infringement of his patent rights has caused him irreparable loss.

Permanent injunction

A permanent injunction is invoked when the case is finally decided by the court. If the defendant is found guilty of patent infringement, the temporary injunction becomes a permanent one. However, the temporary injunction is dissolved and does not become a permanent injunction if the defendant is released from liability.

Damages

In case the defendant is proven guilty, the plaintiff is either awarded damages or an account of profits by the defendant. Damages may not be provided to the plaintiff in case the defendant pleads ignorance and proves that he had no reasonable grounds to believe that the said patent existed at the time of infringement.

2.13: Surrender of patents – Challenges in patents

Surrender of patents

“Sometimes it is wiser to step back and acknowledge than to stand up and confront” – a popular epigram summarizing the ostensible intent behind surrendering a patent (in idiosyncratic circumstances) by a patentee in India. While it might be a strategic decision (or based on a number of other reasons), surrendering a patent, and in turn the rights conferred upon the patentee by the grant of such patent, is an important provision provided in the Indian patents regime.

Cases where the patentee can opt for surrendering his patent

- A patentee wishing to avoid his patent being challenged by an opponent, thereby leading to invalidation of his patent.
- A patentee is unable to pay the fee within the prescribed time
- In any other circumstances like inability to commercialize

Statutory basis and procedure for surrendering the patent

i. Online Notice and application of Surrender of Patents:

Section 63 of the Patents Act, 1970 provides that the patentee may, at any time by giving notice in the prescribed manner to the Controller, offer to surrender his patent. The official fees for providing such an application through online mode is Rs. 1000 /- for natural persons or startups or small entities or educational institutions and Rs. 5000 /- for others as mentioned in the First Schedule of the Patents Rules, 2003.

ii. Publishing the details of surrender

It is further provided in Section 63 that the Controller shall publish the offer for surrendering the patent in the prescribed manner (in the Journal), and also notify every person other than the patentee whose name appears in the register as having an interest in the patent.

iii. Opposition to surrender

If any person interested is aggrieved by the surrender offer, he may within three months from the date of publication of the notice by the Controller, give notice of opposition to the Controller in Form 14 in duplicate. The Controller shall provide an opportunity of being heard to the patentee as well as the opponent

Revoke : If the Controller is satisfied after the hearing that the patent may properly be surrendered, he may accept the offer and, by order, revoke the patent. An important provision while surrendering the patent is that there is no need for the patentee to give the reason or motivation for the offer of the surrender.

Surrendering the patent is a good statutory option available to the patentee to safeguard his financial interest prior to making an offer for the surrender. Any payment made under the patent prior to the date of the acceptance of the surrender application would not be put at risk and surrendering the patent would not adversely affect the continuance of the revocation proceedings in the court. However, the Controller may not proceed with the acceptance of the application till the outcome of the revocation litigation. In case the patent is revoked, it would mean that the patent would be deemed to have never existed and thus subjecting all the financial agreements made earlier at risk.

2.14 : Challenges in Patent/Emerging Issues in Patent

1. Patentability of Software: In the view of emerging trends and innovations in software, the patentability of the software is a challenging issue. Software can be patented only if it is attached to an invention and the software is a component of that invention. Computer programs are not patentable. Hence there is no clarity on patentability of the software.

2. Evergreening of Patents : Evergreening works by filing additional patents related to the original patent. This allows patent holders to extend the life of their inventions and protect them from competitors for longer periods of time. The process typically involves filing multiple follow-up patents that build off of the original patent in order to provide further protection for the invention. Evergreening is common in medicine patents which makes the medicines costlier and inaccessible to public

Examples Of Evergreening Of Patents

Apple Inc. regularly files multiple follow-up patents related to its core products such as iPhones and iPads in order to extend the life of those products' intellectual property protections over an extended period of time.

3. Cost of Patent services: While patent filings for individuals have increased to Rs 1,600 by 60%, those for large firms doubled to Rs 8,000 for every application filed with the Indian Patent Office. The patent agents charge a very higher amount around 1 lakh which prevents inventions

4. Delay in grant of patents: A patent application takes, on average, six years to get approved in India, As many as 98% of patents granted were for applications more than five years old. Patents allow companies with innovative products to benefit from their research and development by giving them exclusive right to make these products, usually for a period of 20 years. If patent grants are delayed, the entry of companies into the Indian market with their product is delayed too.

5. Awareness Issues: Intellectual property is an emerging concept. The people in the developing and less developed countries are not aware of the intellectual property rights and its benefit. Lack of awareness among the public about the procedure and benefits related to intellectual property is a challenging task that need to be addressed.

6. Funding issues : The inventors are not able to find suitable funding platforms for their inventions to be patented. The idea conversion into patented product required financial resources. Very few organizations are available to fund the new inventions which prevents the growth of patents in India.

7. Patent leads to cooperation or competition; Profiting from innovation typically involves a choice between commercializing a patented technology in the product market to exploit proprietary advantage (i.e., competition) or licensing the technology to an incumbent in the market for ideas (a form of cooperation). .